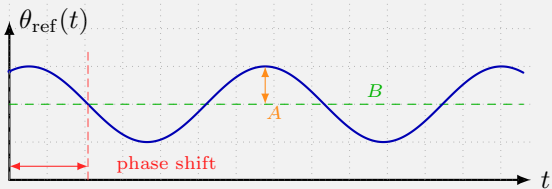


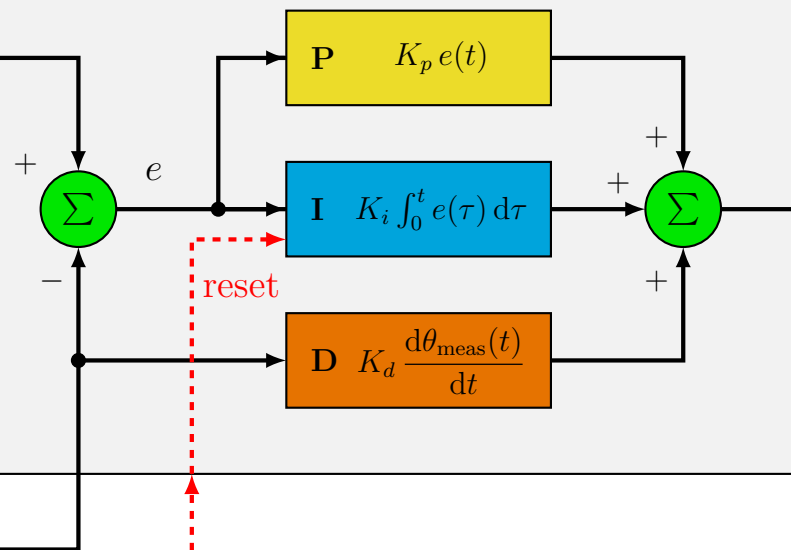
Setpoint



$$\theta_{\text{ref}}(t) = B + A \sin(2\pi f t + \phi)$$

θ_{ref}

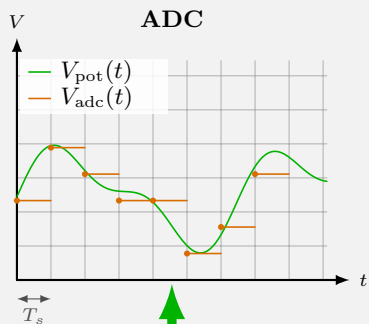
PID Controller



contact event

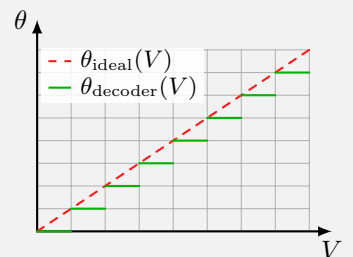
u_{control}

Angle Sensor



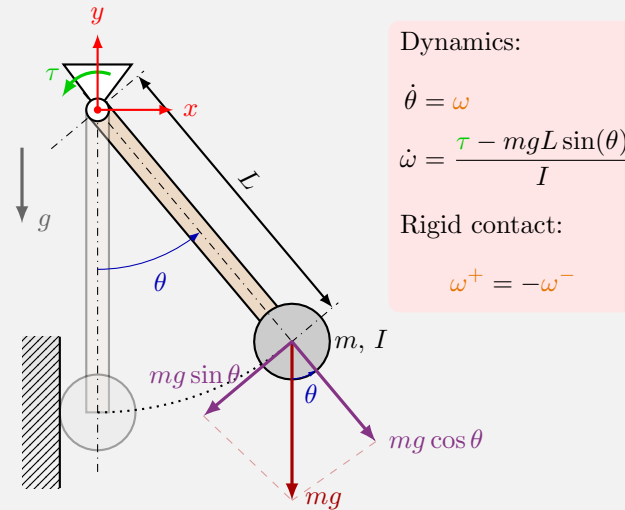
V_{adc}

Decoder



θ_{meas}

Pendulum with Wall Contact



τ

ω

BLDC Drive

